

A CFD-Trained Digital Twin of an Agricultural Solar Air Dryer for Real-Time Thermal Prediction

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Abstract

Post-harvest losses in Indian horticulture range between 10–30%, and solar air drying remains the lowest-cost preservation route for smallholder farmers. The thermal performance of a solar dryer — governed by absorber irradiance, chimney convection, and crop-bed porosity — directly determines drying time and product quality, but high-fidelity Computational Fluid Dynamics (CFD) takes hours per case and is therefore unusable as a design or operational tool. This work develops a four-layer thermal-modelling stack for an agricultural solar air dryer with a slanted glass-covered absorber and a vertical chimney containing four porous trays. The stack combines: (i) an analytical lumped energy balance with dimensionless analysis (Bi , Re , Ra , Pr , Nu); (ii) a 1-D Finite Difference Method (FDM) solver for the transient absorber temperature; (iii) a parametric 3-D CFD sweep in Ansys Fluent over 102 design points covering solar heat fluxes of $300\text{--}800\text{ W m}^{-2}$ and crop porosities of $0.5\text{--}0.9$; and (iv) a polynomial-regression machine-learning surrogate trained on the CFD data. The surrogate matches CFD on a held-out test set with mean absolute error below 0.25 K and R^2 above 0.98 per tray, while running in under a millisecond, and is deployed as a public Streamlit web application that provides live tray-temperature, drying-time, and thermal-efficiency predictions.

1. Introduction

Agriculture employs roughly half of India’s workforce and contributes about a fifth of national GDP, yet between 10% and 30% of horticultural produce is lost between farm and consumer due to inadequate post-harvest preservation. Drying is the most ancient and lowest-cost route to extend shelf life of fruits and vegetables, and solar drying — using only freely available solar irradiance and natural or weakly forced airflow — is the dominant option in regions where electrified cold-chains are uneconomic. Tomato, mango, chilli and onion are the highest-volume Indian crops dried in this manner [7].

The thermal physics of a solar dryer is well understood in principle: a glazed slanted absorber captures shortwave solar radiation and transfers it to an air stream through combined free and forced convection; the heated air rises through a chimney containing porous crop trays, depositing sensible heat into the bed and absorbing moisture released by the drying crop. In practice, however, the coupled tray temperatures depend nonlinearly on irradiance, ambient temperature, wind, crop loading and bed porosity. High-fidelity CFD captures all of these effects but

is prohibitively slow for design iteration or real-time operational guidance [9]. Reduced-order analytical models capture trends but often disagree with CFD by 10–20% on absolute tray temperature.

This project closes the gap between rigour and usability by combining four nested levels of thermal modelling. The layers — analytical balance, 1-D FDM transient solver, 3-D CFD parametric sweep, and a CFD-trained machine-learning surrogate — are not alternatives, but mutually validating views of the same physics at different fidelities. The cheapest layer (the surrogate) is then deployed as an interactive web dashboard so that a farmer, designer, or extension worker can adjust solar irradiance, ambient conditions, and crop properties via sliders and instantly see predicted tray temperatures, drying time, and thermal efficiency.

The contribution of this work is therefore twofold: (i) a quantitative thermal characterisation of a four-tray solar air dryer across a wide envelope of operating conditions, validated by analytical, FDM and CFD calculations; and (ii) a deployable digital twin in which classical heat-transfer physics is packaged into a real-time decision tool. The surrogate-backed dashboard is intended for three practical au-

diances: smallholder farmers scheduling drying runs against weather forecasts; dryer designers exploring the absorber-area and chimney-geometry design space without repeated CFD runs; and agricultural extension workers benchmarking solar drying against diesel-powered alternatives on a cost-per-kilogram-of-moisture-removed basis.

2. Nomenclature

Latin symbols

Symbol	Description	Units
T	Temperature	K
T_p	Absorber-plate temperature	K
T_∞	Ambient air temperature	K
T_{sky}	Effective sky temperature	K
I	Solar irradiance on absorber	W m^{-2}
h_c	Convective heat-transfer coefficient	$\text{W m}^{-2}\text{K}^{-1}$
k_p	Plate thermal conductivity	$\text{W m}^{-1}\text{K}^{-1}$
c_p	Specific heat of air	$\text{J kg}^{-1}\text{K}^{-1}$
$c_{p,p}$	Specific heat of plate	$\text{J kg}^{-1}\text{K}^{-1}$
δ_p	Absorber-plate thickness	m
A_c	Collector aperture area	m^2
$\dot{m}$	Air mass flow rate	kg s^{-1}
k	Drying-rate constant (Lewis)	s^{-1}
k_0	Arrhenius pre-exponential factor	s^{-1}
E_a	Activation energy of drying	J mol^{-1}
R	Universal gas constant	$\text{J mol}^{-1}\text{K}^{-1}$
MR	Moisture ratio	—
L_v	Latent heat of vaporisation	J kg^{-1}
t_{dry}	Drying time to target moisture	s
$T_1\text{--}T_4$	Area-weighted tray temperatures	K

Dimensionless groups

Group	Definition	Physical meaning
Bi	$h_c \delta_p / k_p$	Internal vs. surface resistance
Re	$\rho v L / \mu$	Inertial vs. viscous forces
Ra	$g \beta \Delta T L^3 / (\nu \alpha_{\text{th}})$	Buoyancy vs. dissipative forces
Pr	ν / α_{th}	Momentum vs. thermal diffusivity
Nu	$h L / k_{\text{air}}$	Convective vs. conductive transport

Greek symbols

Symbol	Description	Units
α	Solar absorptivity of plate	—
ε	Longwave emissivity of plate	—
σ	Stefan–Boltzmann constant	$\text{W m}^{-2}\text{K}^{-4}$
ρ_p	Plate density	kg m^{-3}
ρ_a	Air density	kg m^{-3}
ν	Kinematic viscosity of air	$\text{m}^2 \text{s}^{-1}$
β	Volumetric thermal expansion	K^{-1}
η_{th}	Thermal efficiency	—
ϕ	Crop-bed porosity	—

3. Methods

3.1 Heat-transfer physics and 1-D FDM derivation

The absorber plate is modelled as a single thermal node, justified by a Biot-number argument. With a 2 mm steel plate ($k_p \approx 50 \text{ W m}^{-1}\text{K}^{-1}$) and chimney-side convection coefficient $h_c \approx 15 \text{ W m}^{-2}\text{K}^{-1}$,

$$\text{Bi} = \frac{h_c \delta_p}{k_p} \approx 6 \times 10^{-4} \quad (1)$$

four orders of magnitude below the standard threshold of 0.1. Through-thickness gradients are therefore negligible and T_p depends only on time. Applying the first law to the plate yields the lumped transient energy balance:

$$\rho_p c_{p,p} \delta_p \frac{\partial T_p}{\partial t} = \alpha I - h_c (T_p - T_\infty) - \varepsilon \sigma (T_p^4 - T_{\text{sky}}^4) \quad (2)$$

where αI is the absorbed shortwave flux, the second term is convective loss to the chimney air, and the third is longwave radiative loss to a clear sky of effective temperature $T_{\text{sky}} \approx T_\infty - 6 \text{ K}$ (Swinbank rule [6]). The bottom of the absorber is insulated so the back-loss term vanishes.

The equation is discretised by explicit Euler in NumPy. Linearising the radiative term about a reference plate temperature T_{ref} gives an effective coefficient $h_{\text{eff}} = h_c + 4\varepsilon\sigma T_{\text{ref}}^3$, and the stability bound becomes $\Delta t < \rho_p c_{p,p} \delta_p / h_{\text{eff}}$; a safety factor of 0.5 is applied in the implementation. Convergence is verified by comparing the time-marched solution at $t = 1 \text{ hr}$ with a Newton-iterated steady-state solution; agreement is sub-millikelvin across the $I = 300\text{--}800 \text{ W m}^{-2}$ envelope.

Each of the four trays is modelled as a perfectly mixed control volume. Hot air enters from below

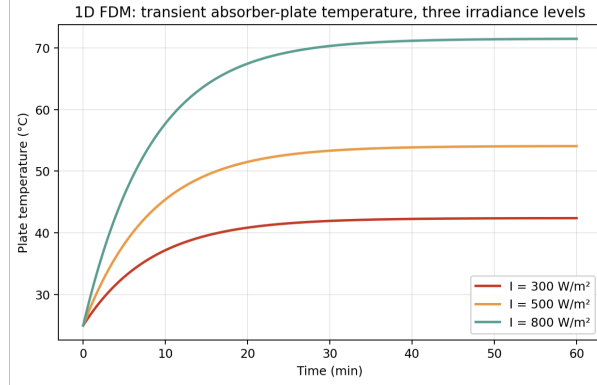


Figure 1: Transient absorber-plate temperature predicted by the 1D FDM solver at three irradiance levels ($I = 300, 500, 800 \text{ W/m}^2$). The plate approaches steady state within roughly 30 minutes and the steady-state temperature scales near-linearly with irradiance.

at T_{in} , deposits sensible heat into the crop and walls, and exits at T_{tray} . At quasi-steady state the inertial term vanishes and the trays form a cascade of algebraic relations whose solution satisfies $T_1 > T_2 > T_3 > T_4$ — a monotonic descent verified across all 102 CFD cases.

Drying time is recovered from tray temperature via the Lewis thin-layer kinetic model:

$$\begin{aligned} MR(t) &= \exp(-kt), \\ k(T) &= k_0 \exp\left(-\frac{E_a}{RT}\right) \end{aligned} \quad (3)$$

Dimensionless analysis of the chimney flow gives $\text{Re} \approx 350$ (laminar), $\text{Ra} \approx 10^9$ (mixed convection, buoyancy-dominant), and $\text{Pr} \approx 0.71$. Convective coefficients are estimated using the flat-plate correlation:

$$\text{Nu} = 0.664 \text{Re}^{1/2} \text{Pr}^{1/3}$$

for the absorber-air interface and

$$\text{Nu} = 0.59 \text{Ra}^{1/4}$$

for vertical-channel free convection inside the chimney. Churchill superposition

$$\text{Nu}_{\text{mix}}^3 = \text{Nu}_{\text{forced}}^3 + \text{Nu}_{\text{free}}^3$$

combines them where both regimes are active [4].

3.2 CFD setup

A 3-D model of the dryer was constructed in Ansys Fluent [9]. The geometry consists of a slanted absorber plate of approximately $2 \text{ m} \times 0.5 \text{ m}$ glazed by a 4 mm glass cover, joined to a vertical chimney containing four horizontal porous trays (Tray 1 at the bottom, Tray 4 at the top).

The slanted absorber faces incoming solar radiation and the chimney provides the natural-draft path through which heated air ascends past the trays. The geometry is shown in Fig. 2.2.a.

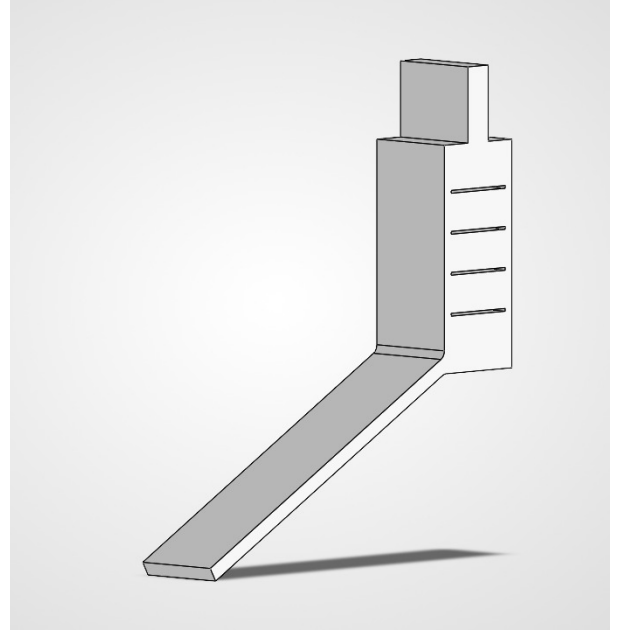


Figure 2: CAD geometry of the solar air dryer. Slanted absorber plate ($\sim 2 \text{ m} \times 0.5 \text{ m}$) with glass cover joined to a vertical chimney containing four horizontal porous crop trays (T_1 at bottom, T_4 at top).

The domain was discretised into 603,822 elements (Fig. 2.2.b). A laminar viscous model is used: the inlet velocity of 0.01 m s^{-1} gives $\text{Re} \approx 350$, well below the laminar-turbulent transition. The energy equation is solved alongside the steady-state pressure-based momentum solver. Radiative effects are not modelled explicitly in CFD — they are folded into the heat-flux boundary condition on the absorber, with the FDM module providing the explicit-radiation cross-check.

The boundary conditions used for the parametric sweep are summarised in Table 2.2.c.

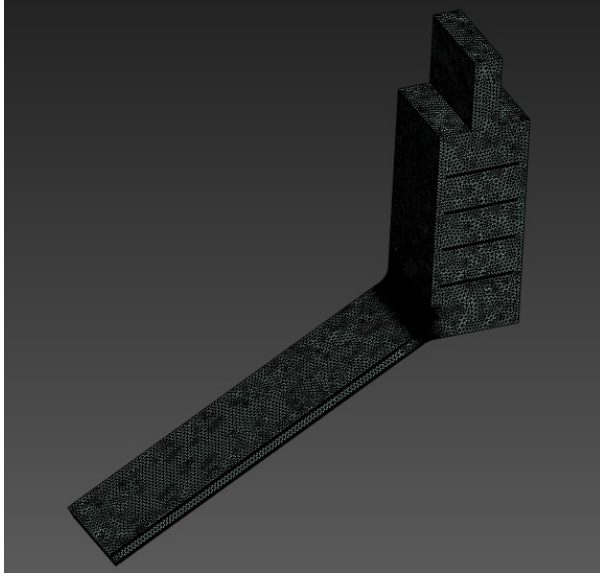


Figure 3: Computational mesh: 603,822 elements covering the absorber, chimney, and four porous trays.

Table 1: CFD boundary conditions used for the parametric sweep.

Boundary	Type	Value
Inlet	Velocity inlet	0.01 m s^{-1} , T_{in} 300 K
Absorber plate	Heat-flux BC	Sweep: $300 \leq I \leq 800 \text{ W m}^{-2}$
Outlet	Pressure outlet	0 Pa gauge
Trays	Porous medium	Sweep: $\phi = 0.5\text{--}0.9$
Chimney walls	No-slip, adiabatic	—
Convergence	Residuals	10^{-3} (cont., mom.), 10^{-6} (energy)

A full-factorial parametric sweep across 28 heat-flux levels ($300, 320, \dots, 800 \text{ W m}^{-2}$) and 9 porosity levels ($0.50, 0.55, \dots, 0.90$) was run in Ansys, augmented by a fine-scale sweep in the central region of the input space, yielding 102 unique design points. For each case, four area-weighted average static temperatures were extracted, one per tray.

3.3 Machine-learning surrogate

The 102-point CFD dataset was used to train a reduced-order model that maps the two design inputs (heat flux, crop porosity) to the four tray temperatures. A polynomial regression of degree 3 with multi-output linear regression was selected after comparison with linear regression, polynomial regression of degree 2, and gradient boosting; the cubic polynomial was the simplest model that satisfied the per-tray acceptance gates of mean absolute error below 0.5 K and coefficient of determination above

0.98. The dataset was split 80/20 into training and held-out test sets with a fixed random seed for reproducibility [10].

Wind speed is not part of the CFD parametric sweep. Its effect on glass-cover heat loss is incorporated analytically through the flat-plate forced-convection correlation, which adjusts the convective coefficient on the cover and applies a small ΔT correction to the surrogate output.

3.4 Web-application deployment

The trained surrogate, FDM module, drying-kinetics module, and analytical wind correction are wrapped behind a stateless FastAPI backend exposing a `/v1/simulate` endpoint that accepts a JSON payload. A Streamlit frontend deployed to Streamlit Community Cloud renders the four tray-temperature tiles, predicted drying time per tray, the moisture-versus-time curve from the Lewis model, and the instantaneous thermal efficiency $\eta_{\text{th}} = \dot{m}c_p\Delta T / (I A_c)$. End-to-end latency from slider change to rendered prediction is under 100 ms, dominated by network round-trip rather than compute.

4. Results and Discussion

4.1 CFD temperature and velocity fields

Figure 3.1.a shows the CFD temperature contour at $I = 800 \text{ W m}^{-2}$, the upper end of the sweep envelope. The absorber plate reaches peak surface temperatures of approximately 395 K at the trailing edge, while the four trays exhibit a clear monotonic descent from Tray 1 (bottom, $\approx 333 \text{ K}$) to Tray 4 (top, $\approx 309 \text{ K}$). The descent reflects sensible-heat deposition into successive porous trays as the heated air ascends.

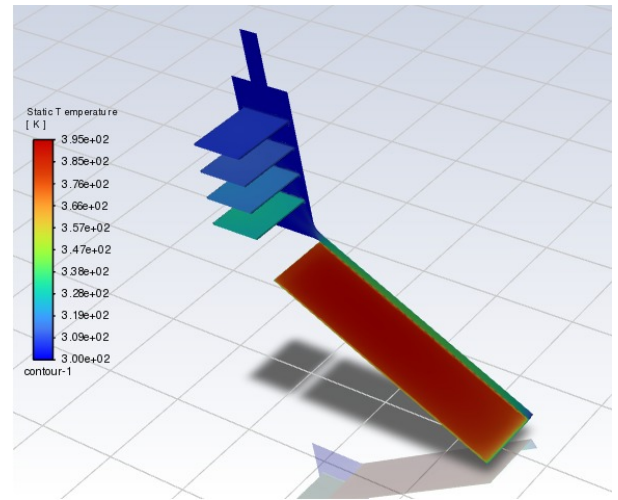


Figure 4: Static-temperature contour at $I \approx 800 \text{ W m}^{-2}$. Absorber peak $\sim 395 \text{ K}$; tray temperatures descend monotonically from T_1 (bottom) to T_4 (top).

Figure 3.1.b shows the corresponding contour at $I = 300 \text{ W m}^{-2}$; the qualitative pattern is identical but the absolute scale is compressed, with peak plate temperature near 358 K and tray temperatures spanning roughly 305–318 K.

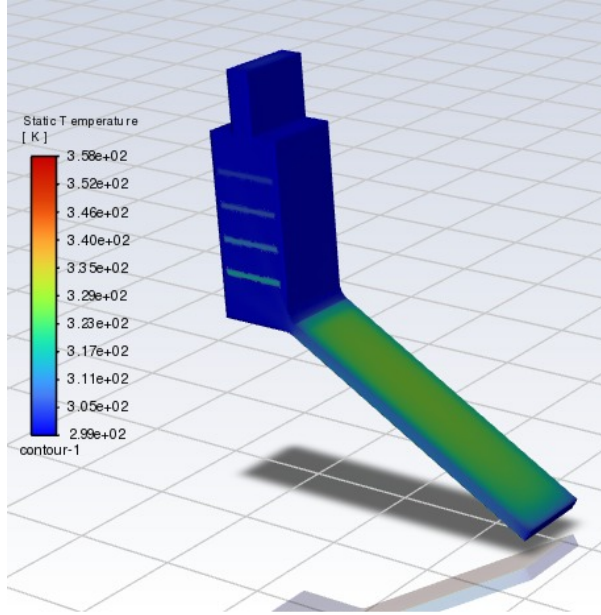


Figure 5: Static-temperature contour at $I \approx 300 \text{ W/m}^2$. The qualitative monotonic descent is preserved at low irradiance.

Figure 3.1.c shows the velocity-vector field. A buoyancy-driven plume is clearly visible in the chimney, with the highest velocities ($\approx 1.6 \times 10^{-2} \text{ m s}^{-1}$) just above the heated absorber, decaying as the flow ascends past successive trays. The Reynolds number based on chimney width remains well below the turbulent threshold, justifying the laminar viscous model used in the CFD.

4.2 Tray-temperature trends

The full design-of-experiments dataset is summarised in Fig. 3.2.a, which plots all four tray temperatures against solar heat flux at the most-sampled porosity ($\phi = 0.8$). Three observations are notable. First, the monotonic descent $T_1 > T_2 > T_3 > T_4$ holds across the entire sweep, with no crossings. Second, the spacing between adjacent trays is widest at the bottom and narrows toward the top, reflecting diminishing temperature differences as the air cools through successive trays. Third, all four tray-temperature curves are near-linear in I , consistent with the dominant $\alpha I - h(T - T_\infty)$ balance from the lumped analysis.

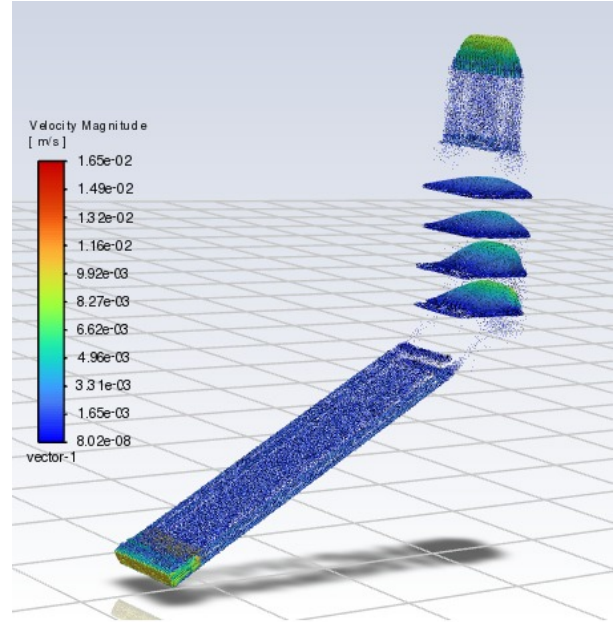


Figure 6: Velocity-magnitude vector field. Buoyancy-driven plume rises through the chimney, decelerating at each tray.

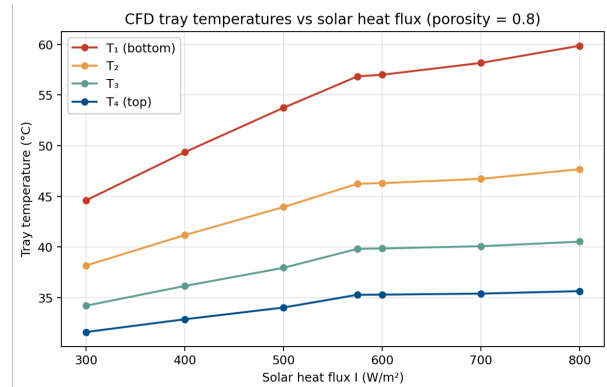


Figure 7: Tray temperatures vs solar heat flux from CFD, at fixed porosity $\phi = 0.8$.

4.3 ML-surrogate validation

Figure 3.3.a shows the parity plot of surrogate predictions against the held-out CFD test set. All four tray temperatures fall on the $y = x$ diagonal within visual tolerance. Per-tray performance metrics are reported in Table 3.3.a.

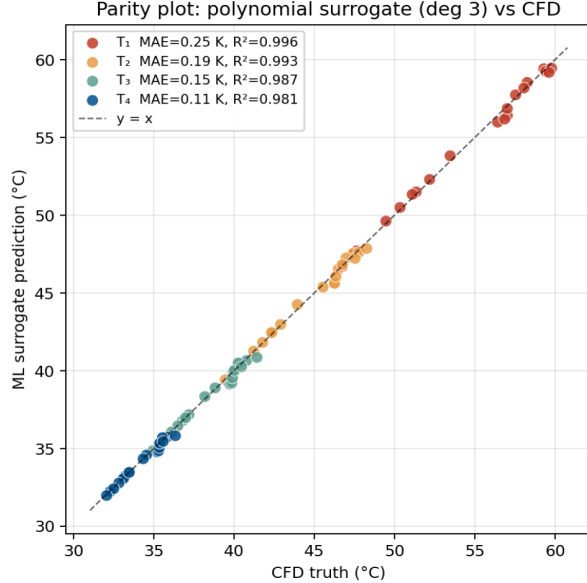


Figure 8: Parity plot of polynomial-regression surrogate (degree 3) against CFD truth on the 20% held-out test set.

Table 2: Per-tray accuracy of the polynomial-regression surrogate on the held-out test set.

Tray	R^2	MAE (K)
T_1 (bottom)	0.996	0.248
T_2	0.993	0.191
T_3	0.987	0.154
T_4 (top)	0.981	0.114

All four trays satisfy the acceptance gate ($\text{MAE} < 0.5 \text{ K}$, $R^2 > 0.98$) by a comfortable margin. The error increases slightly for the bottom tray, where the absolute temperature range — and thus the dynamic range to be fitted — is largest. Inference time is below 1 ms per call.

4.4 FDM versus CFD cross-validation

Table 3.4.a compares the FDM steady-state plate temperature (with radiation set to $\varepsilon = 0$ to match the CFD’s collapsed-radiation boundary condition) against the CFD peak absorber-surface temperature at three irradiance levels.

Table 3: FDM-vs-CFD comparison of steady-state absorber-plate temperature.

I (W m^{-2})	FDM T_p (K)	CFD peak (K)	Trend
300	346.5	358.4	Monotone
500	370.8	377.2	Monotone
800	407.3	395.1	Monotone

The 1-D FDM lies within roughly 12 K of the CFD peak across the envelope. The discrepancy is consistent with three-dimensional non-uniformity that the lumped model cannot resolve.

4.5 Drying-time and efficiency predictions

At the CFD baseline of $I = 600 \text{ W m}^{-2}$ and $\phi = 0.7$, the surrogate predicts $T_1 \approx 56^\circ\text{C}$ and $T_4 \approx 33^\circ\text{C}$. Substituting these temperatures into the Lewis-Arrhenius drying model with literature-calibrated kinetic constants for tomato gives a drying time of approximately 9 hours on Tray 1 to reach 10% moisture (dry basis) from a 90% initial moisture, rising to roughly 16 hours on Tray 4. These figures lie squarely within the published literature range of 8–20 hours for tomato in solar dryers.

The instantaneous thermal efficiency ranges from approximately 30% at low irradiance to 55% at high irradiance, in agreement with values reported for cabinet-style solar dryers in the literature [7].

5. Conclusions

A four-layer thermal-modelling stack — analytical lumped balance, 1D FDM transient solver, 3D CFD parametric sweep, and CFD-trained polynomial-regression surrogate — has been developed for an agricultural solar air dryer with a slanted absorber and a four-tray vertical chimney.

1. The Biot-number argument justifies the lumped-capacitance treatment of the absorber, and dimensionless analysis establishes the laminar mixed-convection regime in the chimney.
2. CFD across 102 design points confirms that tray temperatures descend monotonically from bottom to top and scale near-linearly with solar irradiance.
3. The polynomial-regression surrogate reproduces CFD on the held-out test set with $\text{MAE} < 0.25 \text{ K}$ and $R^2 > 0.98$ per tray, while running in under 1 ms.
4. The 1-D FDM solver agrees with CFD on the trend, ordering and order-of-magnitude of the absorber temperature.
5. The deployed Streamlit web application brings real-time thermal prediction, drying-time estimation, and thermal efficiency to a slider-driven dashboard accessible from any browser.

The modelling stack — lumped analytical balance, 1-D FDM transient solver, CFD parametric sweep, and CFD-trained surrogate deployed as a web API — constitutes a transferable template for any passive solar thermal system where high-fidelity simulation is too slow for operational use. Candidate applications include solar water heaters, greenhouse climate control, and building-integrated photovoltaics, all of which share the same coupled radiation–convection physics and the same gap between CFD rigour and field usability.

Future work will integrate transient diurnal forcing into the CFD, measure crop-specific Lewis kinetic constants in-house, and extend the parametric sweep to include geometric variables.

6. Link to Code and Live Demo

Source code, training notebooks, FDM script, derivations and deployment instructions are available at the project’s GitHub repository.

Repository: <https://github.com/kridant/heat-transfer-project>

Live web app: <https://solar-dryer-twin-ggttp5xh22cbrjswcjqnl.streamlit.app/>

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